

GREENBRIER VALLEY, WV, USA

WMO: 724127

Lat: 37.867N

Lon: 80.400W

Elev: 2302

StdP: 13.51

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 53801

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	11.6	-3.5	5.0	12.3	1.2	6.4	16.6	18.6	35.3	16.7	32.9	4.8	300	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	87.1	69.4	84.1	68.8	81.9	68.0	73.4	80.7	72.1	79.4	70.9	78.0	5.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	128.2	77.2	70.2	120.9	75.9	68.5	114.1	74.4	38.6	81.2	37.4	79.1	36.3	77.7	95.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	12.9	11.4	DB	0.4	90.7	5.8	7.7	-3.7	96.3	-7.1	100.8	-10.4	105.2	-14.6	110.8	
			WB	-0.1	75.9	5.7	5.2	-4.3	79.6	-7.6	82.7	-10.8	85.6	-15.0	89.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	30.4	33.4	41.3	52.2	60.0	66.9	70.2	69.5	63.8	52.7	42.7	34.1
	DBStd	16.37	10.93	9.94	10.37	8.93	7.88	5.39	4.83	4.86	6.61	8.53	8.69	9.64
	HDD50	2312	611	473	309	80	14	0	0	0	3	71	251	500
	HDD65	5459	1072	886	736	387	191	43	13	17	97	390	670	957
	CDD50	2865	4	7	38	147	324	506	625	605	416	154	31	8
	CDD65	536	0	0	0	4	35	99	173	157	60	8	0	0
	CDH74	4186	0	0	9	106	346	767	1254	1119	506	78	1	0
	CDH80	946	0	0	0	12	51	151	327	275	118	12	0	0
(14) Wind	WSAvg	3.9	5.0	5.0	5.0	5.0	3.8	3.3	2.9	2.4	2.6	3.3	3.7	4.4
(15) Precipitation	PrecAvg	38.2	2.8	2.7	3.4	3.3	3.8	3.4	4.1	3.4	2.9	2.6	2.7	2.9
	PrecMax	57.1	6.9	6.6	7.8	7.1	8.0	8.1	9.1	7.8	6.9	6.3	7.3	6.1
	PrecMin	26.4	0.4	0.6	0.4	0.5	0.9	0.8	1.8	0.9	0.0	0.0	0.3	0.5
	PrecStd	6.5	1.3	1.2	1.7	1.5	1.8	1.8	1.7	1.7	1.8	1.5	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	66.0	74.6	81.6	84.3	88.0	90.4	90.1	88.2	81.0	71.8	64.4
		MCWB	55.6	56.3	57.1	58.3	66.0	69.8	70.4	71.6	66.7	65.3	60.3	59.1
	2%	DB	56.0	59.3	69.3	77.1	81.2	84.0	86.4	86.0	83.6	75.3	65.9	58.3
		MCWB	51.6	51.7	55.8	59.1	64.9	68.7	70.2	69.3	67.0	62.4	56.0	54.0
	5%	DB	51.8	54.3	63.5	72.8	78.4	81.5	83.8	82.4	80.5	72.0	62.5	53.8
		MCWB	48.2	47.1	52.4	57.1	64.0	67.5	69.6	69.3	66.5	60.9	54.6	49.7
	10%	DB	46.0	49.7	59.0	69.5	74.6	79.0	81.3	80.9	76.2	67.7	57.3	49.0
		MCWB	42.8	43.8	50.3	56.4	63.0	66.9	69.4	68.8	64.9	60.1	51.4	45.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	59.1	60.5	65.3	69.5	72.9	75.1	75.4	72.2	69.0	62.6	60.4
		MCDB	60.6	64.3	70.4	75.4	79.1	81.4	82.2	83.5	80.1	77.0	66.5	64.2
	2%	WB	53.2	53.7	57.4	62.6	68.1	71.3	73.5	73.5	70.2	66.6	59.2	55.0
		MCDB	55.4	57.5	65.3	70.9	76.8	79.3	80.9	80.6	77.8	72.5	64.5	57.8
	5%	WB	48.0	48.3	55.0	60.2	66.5	70.1	72.3	72.0	68.7	63.6	56.0	50.4
		MCDB	50.7	52.8	61.8	68.0	74.5	77.9	79.6	78.9	75.8	67.6	60.2	52.9
	10%	WB	43.1	43.7	51.6	57.7	64.5	68.8	71.1	70.4	67.0	60.6	52.8	45.9
		MCDB	46.1	48.3	56.8	65.7	71.4	76.2	78.2	77.0	73.6	65.8	56.1	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.7	20.1	23.2	21.2	20.3	19.2	19.7	20.9	21.2	20.6	16.1
		MCDBR	22.3	25.7	28.1	30.2	26.1	23.5	22.7	22.9	26.1	27.4	27.8	22.2
		MCWBR	17.5	18.1	16.2	14.7	12.0	10.5	9.1	9.5	11.4	14.7	18.3	16.9
	5% WB	MCDBR	19.9	22.0	24.5	23.3	22.1	20.4	19.5	19.3	20.7	20.5	21.3	20.1
		MCWBR	17.0	17.6	15.9	13.2	11.7	10.7	10.2	10.0	10.4	12.6	16.4	17.3
(40) Clear-Sky Solar Irradiance	taub		0.292	0.307	0.331	0.370	0.409	0.433	0.462	0.468	0.369	0.342	0.307	0.290
	taud		2.496	2.440	2.432	2.351	2.244	2.206	2.153	2.109	2.384	2.459	2.529	2.532
	Ebn at Noon		285	294	296	288	277	269	260	256	279	277	277	279
	Edh at Noon		25	30	34	39	44	45	48	49	35	29	24	23
(44) All-Sky Solar Radiation	RadAvg		635	865	1202	1554	1722	1896	1792	1648	1383	1053	768	558
	RadStd		63	111	82	124	148	128	108	84	144	131	83	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (32 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+113

Nomenclature: See separate page